

Machine Learning - Test Notes

Machine Learning is a branch of Artificial Intelligence that enables computers to learn from data and improve performance without being explicitly programmed.

Supervised Learning

Supervised learning uses labeled data to train models. Common algorithms include Linear Regression, Logistic Regression, Decision Trees, and Support Vector Machines.

Unsupervised Learning

Unsupervised learning discovers hidden patterns in unlabeled data. Examples include Clustering and Association Rule Mining.

Applications of Machine Learning

Machine Learning is widely used in retail, banking, healthcare, fraud detection, recommendation systems, image recognition, and natural language processing.

Data Mining

Application of machine learning methods to large databases is called Data Mining. It helps discover patterns, trends, and useful insights from data.

Advantages

Automation, predictive analytics, improved decision making, and scalability.

Disadvantages

Requires quality data, computational resources, and may suffer from bias.